BOE-601	Total Quality Management	L T P 3 0 0	3 Credits

Pre-requisites: None

Course Objectives:

CO1	Aware about the total quality management concept.
CO2	Control Quality in the critical activities of an organization by bringing together resources, equipment, people and procedures.
CO3	Learn to construction and analysis of R charts
CO4	Learn to construction and analysis of C charts.
CO5	To calculate reliability, building reliability in the product

Detailed Syllabus

MODULE 1

Quality Concepts:
Evolution of Quality Control, concept change, TQM Modern concept, Quality concept in design, Review of design, Evolution of proto type.

Control on Purchased Product
Procurement of various products, evaluation of supplies, capacity verification, Development of sources, procurement procedure.

Manufacturing Quality
Methods and techniques for manufacture, inspection and control of product, quality in sales and services, guarantee, analysis of claims.

MODUEL II

Quality Management
Organization structure and design, quality function, decentralization, designing and fitting, organization for different type products and company, economics of quality value and contribution, quality cost, optimizing quality cost, seduction program.

Control Charts
Theory of control charts, measurement range, construction and analysis of R charts, process capability study, use of control charts.

Attributes of Control Chart
Defects, construction and analysis of charts, improvement by control chart, variable sample size, construction and analysis of C charts.

MODUEL III

Defects diagnosis and prevention defect study, identification and analysis of defects, correcting measure, factors affecting reliability, MTTF, calculation of reliability, building reliability in the product, evaluation of reliability, interpretation of test results, reliability control, maintainability, zero defects, quality circle. **,**ISO-9000 and its concept of Quality Management, SO 9000 series, Taguchi method, JIT in some details. **7**

Text Books:

1. Lt. Gen. H. Lal, "Total Quality Management", Eastern Limited, 1990.

2. Greg Bounds, "Beyond Total Quality Management", McGraw Hill, 1994.

3. Menon, H.G, "TQM in New Product manufacturing", McGraw Hill 1992.

Course Outcomes: After the completion of the course the student will be able to:

CO1	To realize the importance of significance of quality
CO2	Manage quality improvement teams
CO3	Identify requirements of quality improvement programs
CO4	To have exposure to challenges in Quality Improvement Programs
CO5	To have a good understanding of the concept of Quality